The Mathematics of a Trading System

Expectancy, variance, the risk/reward tradeoff and the risk of ruin — measured, not assumed, on real OHLCV.

A walk through what a rules-based backtester actually proves once costs, position sizing and luck are taken seriously. It runs two studies through the same engine: an **underpowered crossover** on 5 instruments (SPY, QQQ, PETR4.SA, VALE3.SA, ITUB4.SA), where ~40 trades each cannot resolve the question; and a **powered RSI(2) mean-reversion** study over 20 instruments and **2,313 trades**, where it finally can — and the honest answer, after costs, is no edge.

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1. Method — where the numbers come from

The win rate, expectancy and risk figures in this report are **not downloaded**. Yahoo Finance supplies only OHLCV; everything else is a *result* of simulating the strategy trade by trade. The engine is built to avoid the errors that flatter a backtest:

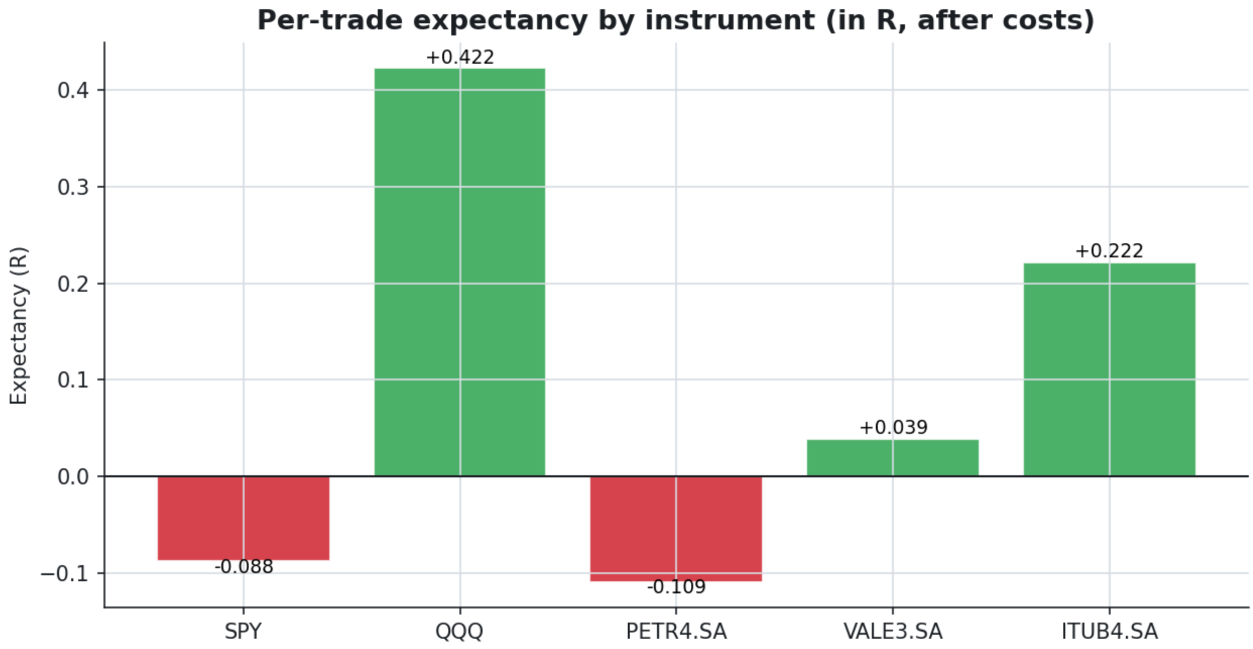
• **No lookahead.** A signal at the close of bar t is executed at the open of bar t+1 — never at the price that produced it.
• **Costs on every trade.** Spread, commission and slippage are charged round-trip; without them, paper expectancy lies.
• **Fixed-fractional risk.** Each trade risks a constant percentage of equity; a wider stop means a smaller position.
• **Conservative fills.** If a bar touches both stop and target, the stop is assumed to fill first.
• **Reproducible variance.** The Monte-Carlo bootstrap uses a fixed seed, so every run is identical.

Result in R (the R-multiple) is the unit that connects everything: a system is described by its win rate and its average R. A loss at the stop is –1R; the breakeven win rate, 1/(R+1), is the bar the real win rate must clear.

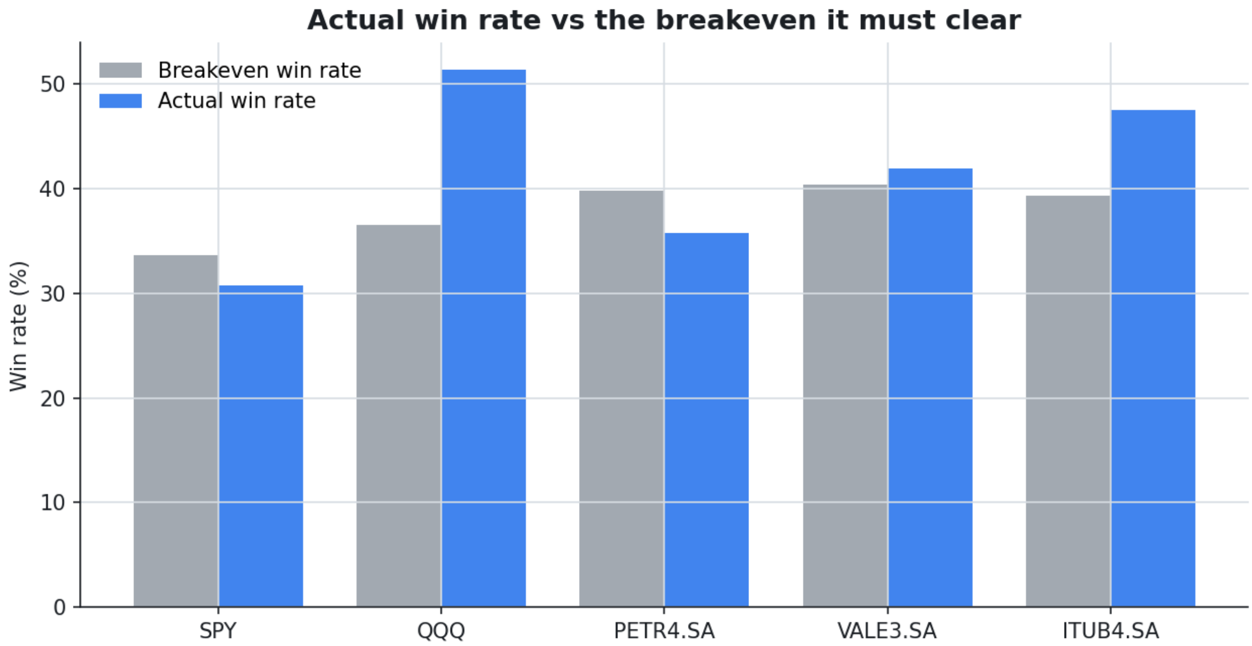
2. Cross-instrument scorecard

The same strategy, same parameters, run on every instrument. The point estimates differ in sign — but the "Verdict" column withholds judgement, because §3 shows none of them is distinguishable from zero at ~40 trades. Read the numbers as hypotheses, not findings.

Instrument	Trades	Win%	Exp. R	Exp. $	Profit factor	Max DD%	Verdict
SPY	39	30.8	-0.088	-4.61	0.87	5.6	unconfirmed
QQQ	37	51.4	+0.422	+21.63	1.83	1.9	unconfirmed
PETR4.SA	42	35.7	-0.109	-5.63	0.84	3.7	unconfirmed
VALE3.SA	43	41.9	+0.039	+1.70	1.06	3.6	unconfirmed
ITUB4.SA	40	47.5	+0.222	+11.09	1.40	4.7	unconfirmed



Per-trade point estimates in R, after costs — not yet confidence-checked (see §3).



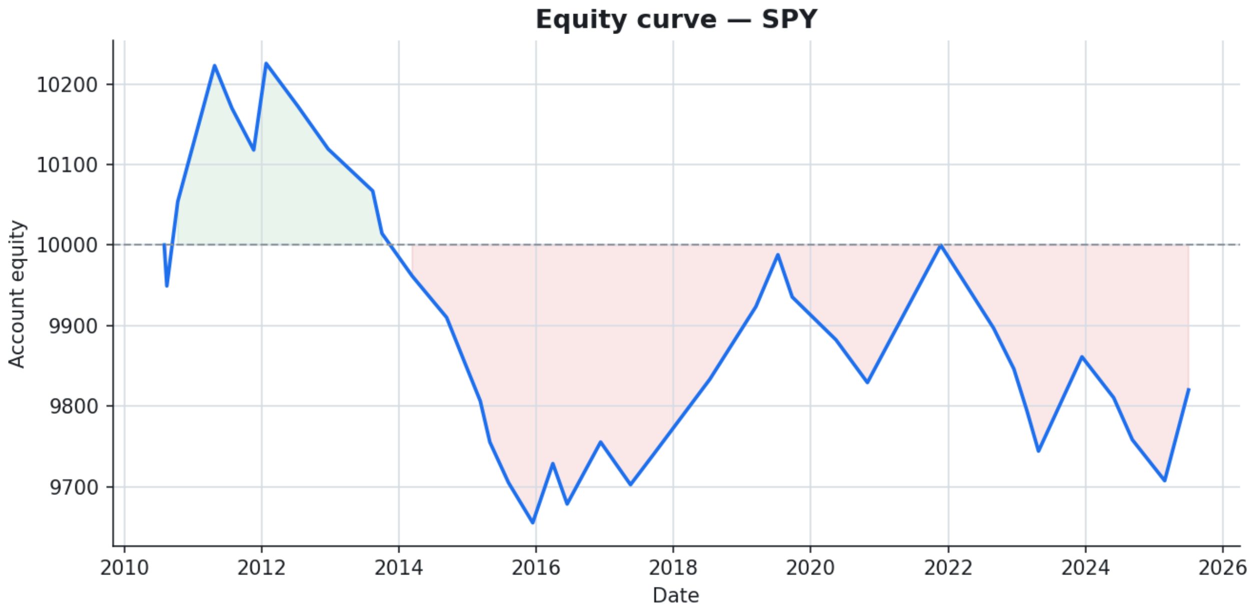
Actual win rate against the breakeven it must beat. A bar to the right of its grey twin is profitable.

But none of these point estimates is established. With ~40 trades each, the spread of expectancies is exactly what pure chance produces even if the true edge were identical (or zero) everywhere. The next section tests that honestly.

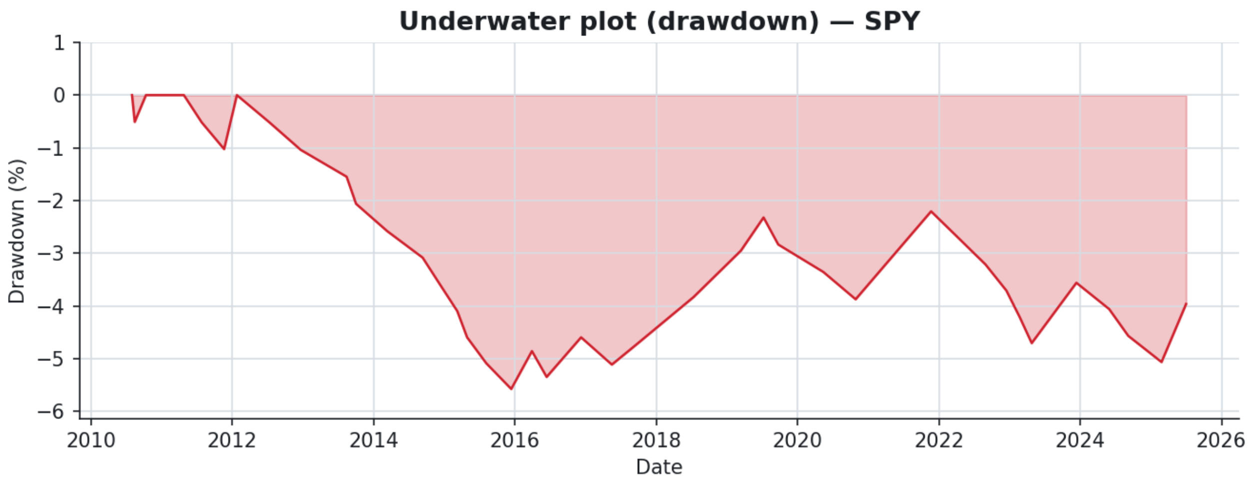
3. SPY — the full fingerprint

Metric	Value	Metric	Value
Trades	39	**Profit factor**	0.87
Win rate	30.77%	**Payoff (R)**	1.98
Avg win	101.45 (+2.06R)	**Avg loss**	-51.76 (-1.04R)
Expectancy $	-4.61	**Expectancy R**	-0.088
Breakeven WR	33.59%	**Actual WR**	30.77%
Total return	-1.80%	**CAGR**	-0.12%
Max drawdown	5.58%	**Sharpe / Sortino**	-0.10 / -6.90

■ Small sample (39 trades < 100): expectancy is still noise-dominated. Treat with skepticism. In particular, **Sharpe / Sortino and the risk-of-ruin percentages are unstable at this n** — their decimals carry false precision and should be read as rough indications only; the significance and pooling sections (§3-4) are where the conclusions live.



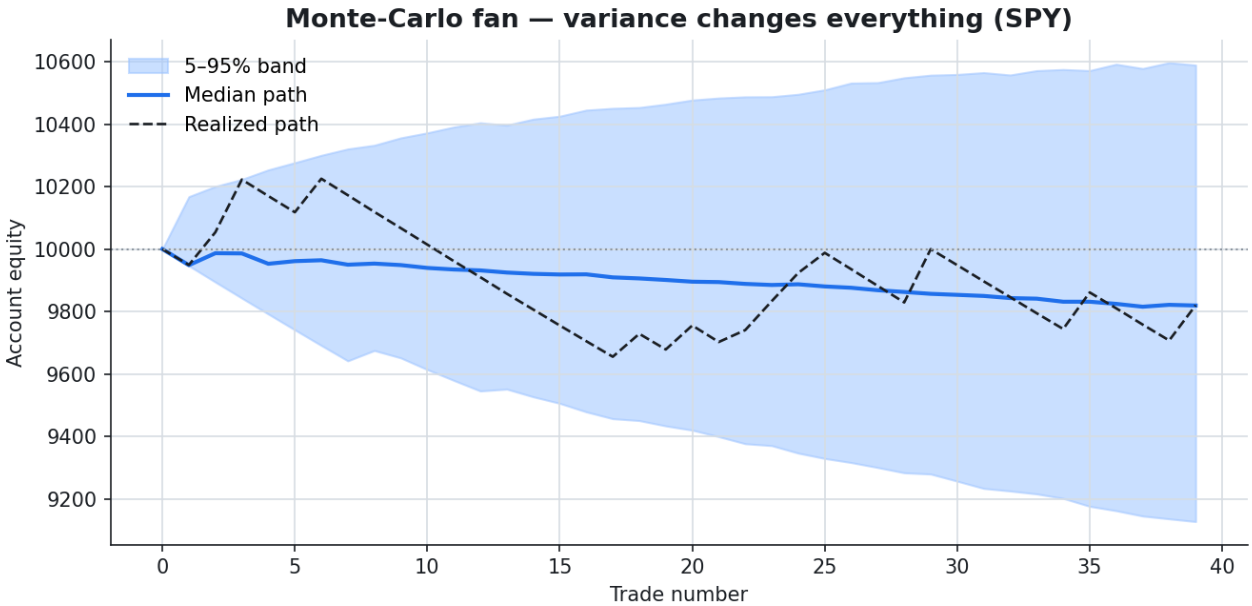
Realized equity curve, costs included.



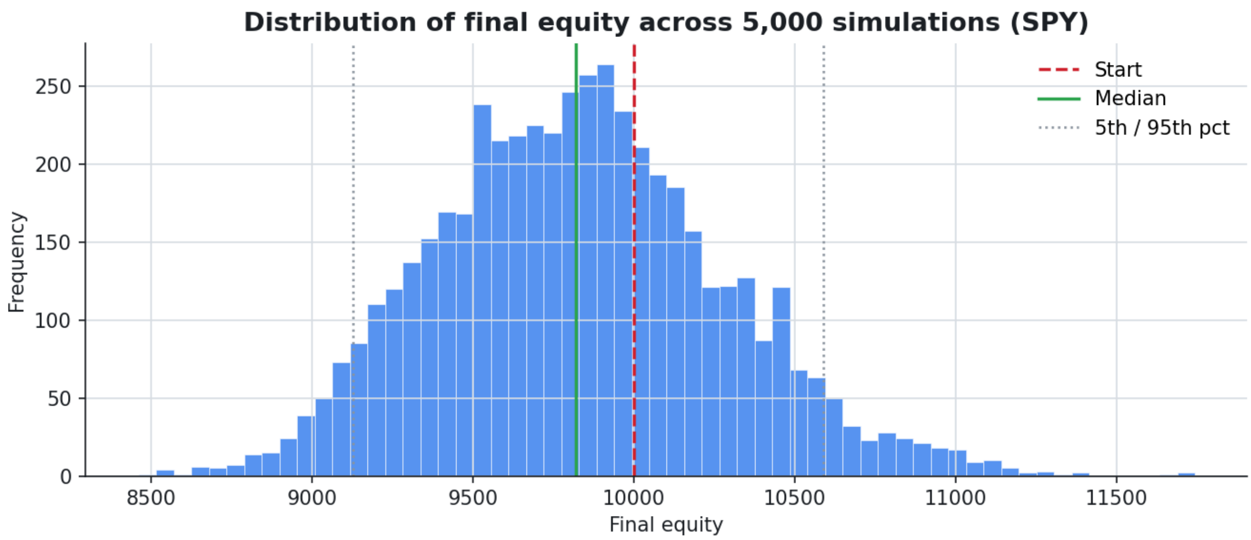
Underwater plot: time spent below the previous equity peak.

3.1 Variance — same edge, different luck

Resampling the realized trades *with replacement* 5,000 times shows the range of outcomes the same system could have produced. (A plain reshuffle would leave the final equity unchanged under fixed-fractional sizing — a product commutes — so resampling with replacement is what disperses the destination, which is exactly what the histogram shows.) The median is the honest expectation; the 5–95% band is the experience you must be able to sit through.

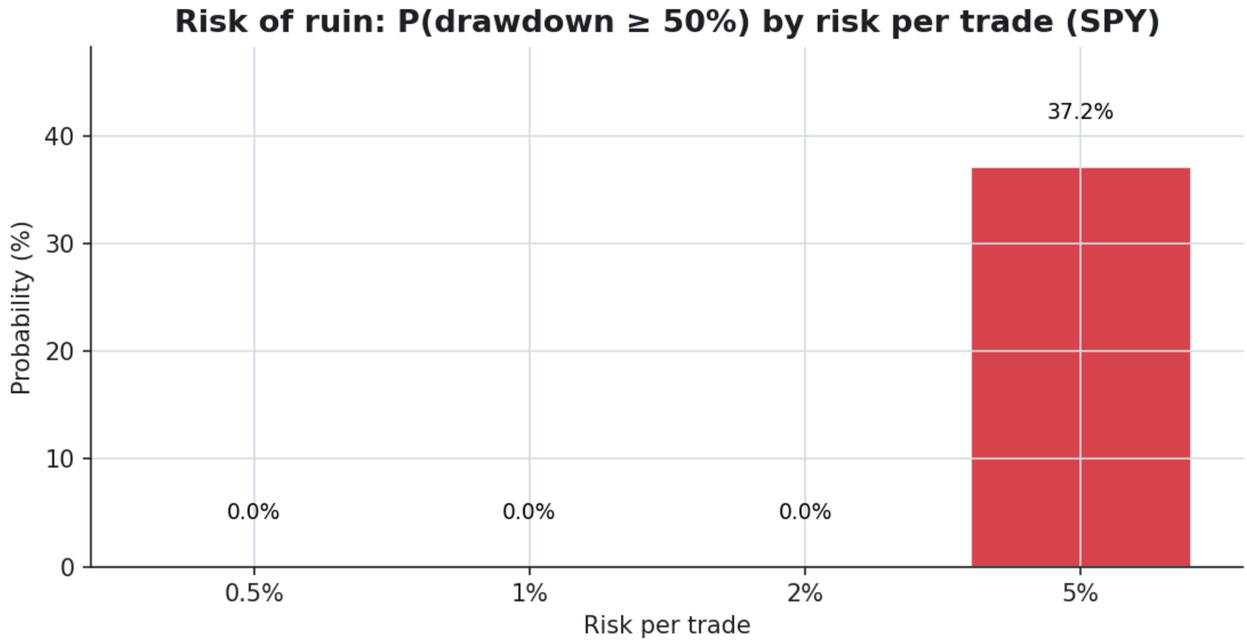


Monte-Carlo fan: median path and 5–95% band vs the single realized run.

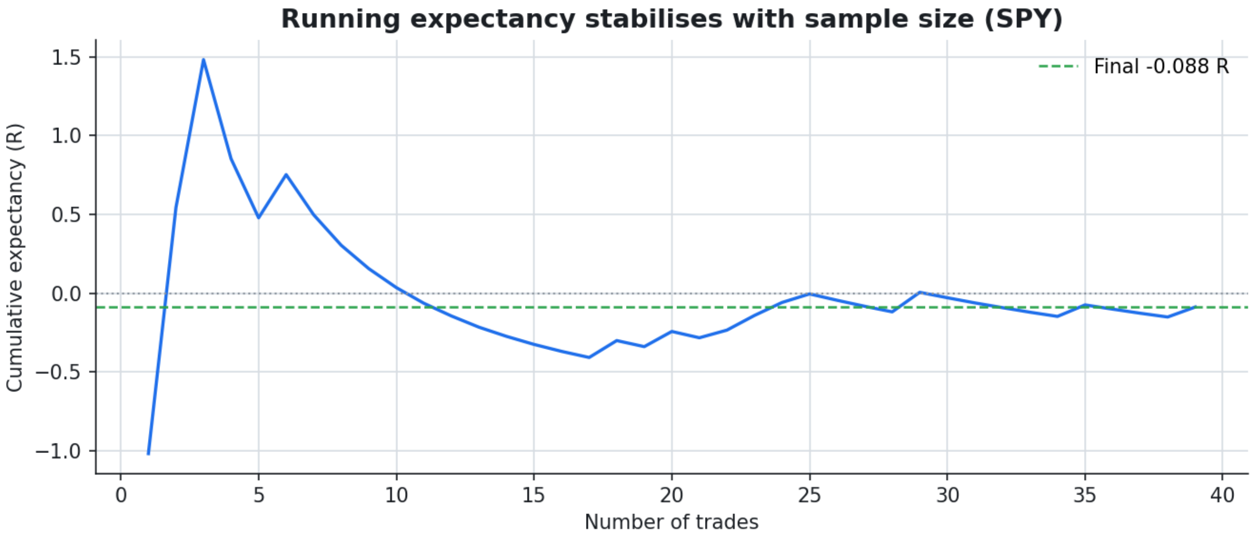


Distribution of final equity across the simulations.

3.2 Risk of ruin & sample size



Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.

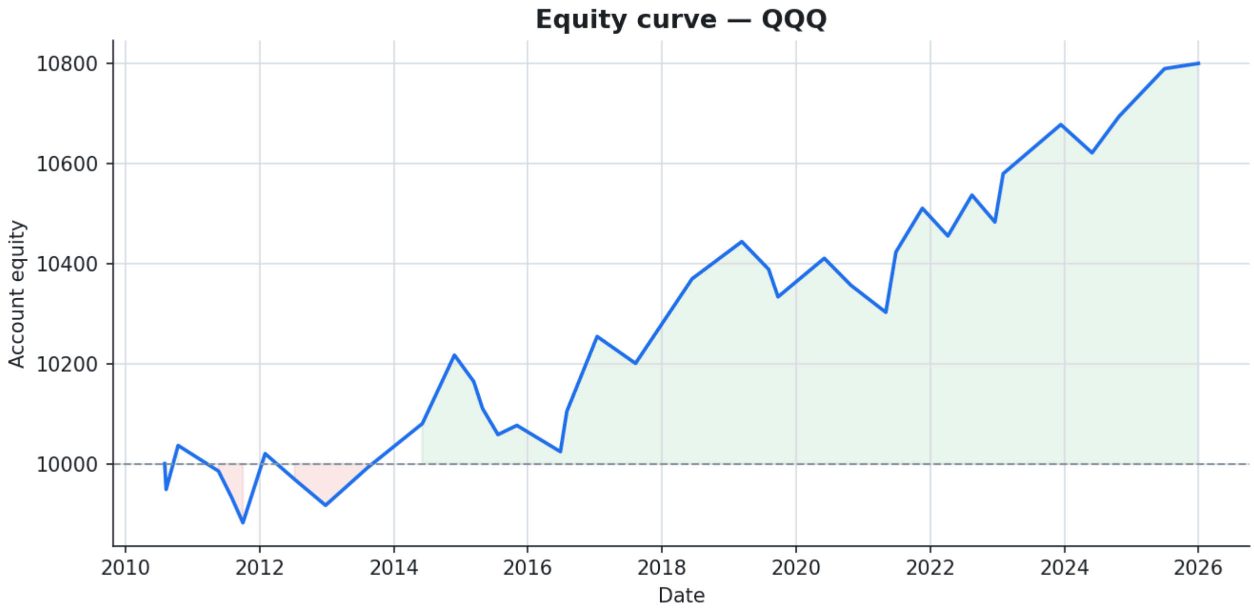


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

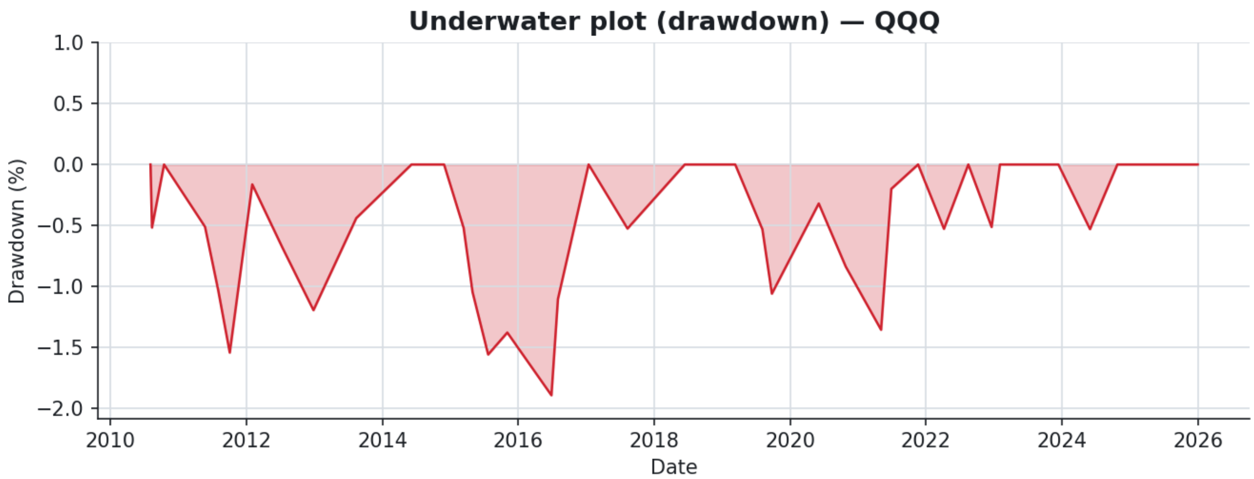
4. QQQ — the full fingerprint

Metric	Value	Metric	Value
Trades	37	**Profit factor**	1.83
Win rate	51.35%	**Payoff (R)**	1.74
Avg win	92.63 (+1.81R)	**Avg loss**	-53.31 (-1.04R)
Expectancy $	21.63	**Expectancy R**	+0.422
Breakeven WR	36.54%	**Actual WR**	51.35%
Total return	+8.00%	**CAGR**	+0.50%
Max drawdown	1.89%	**Sharpe / Sortino**	0.42 / 20.79

■ Small sample (37 trades < 100): expectancy is still noise-dominated. Treat with skepticism. In particular, **Sharpe / Sortino and the risk-of-ruin percentages are unstable at this n** — their decimals carry false precision and should be read as rough indications only; the significance and pooling sections (§3-4) are where the conclusions live.



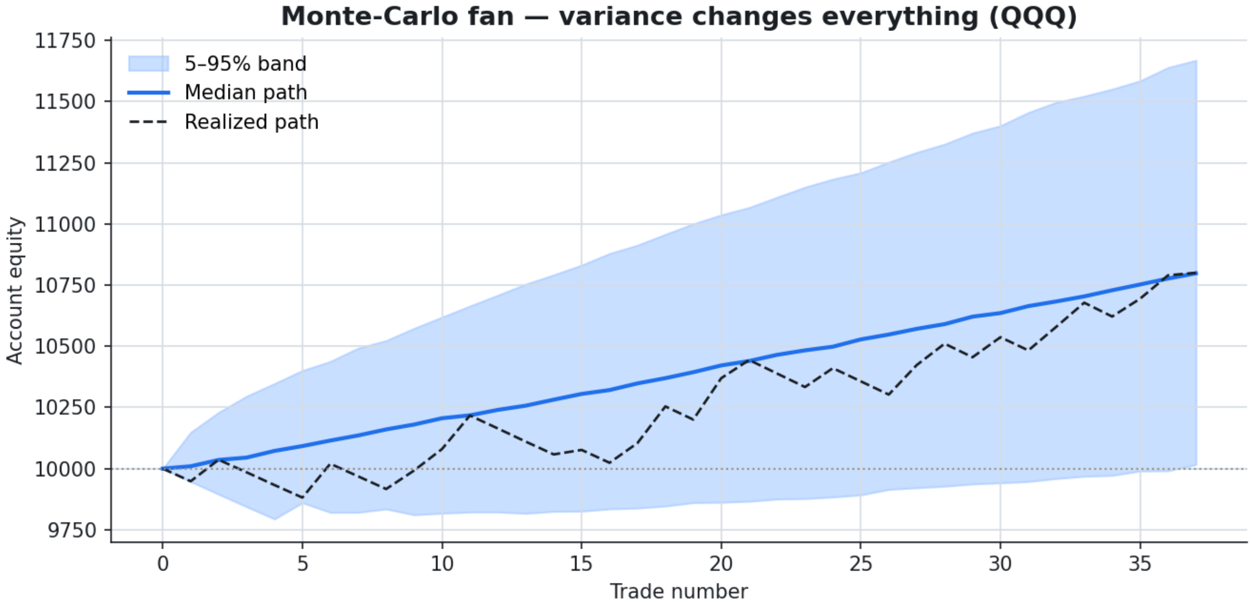
Realized equity curve, costs included.



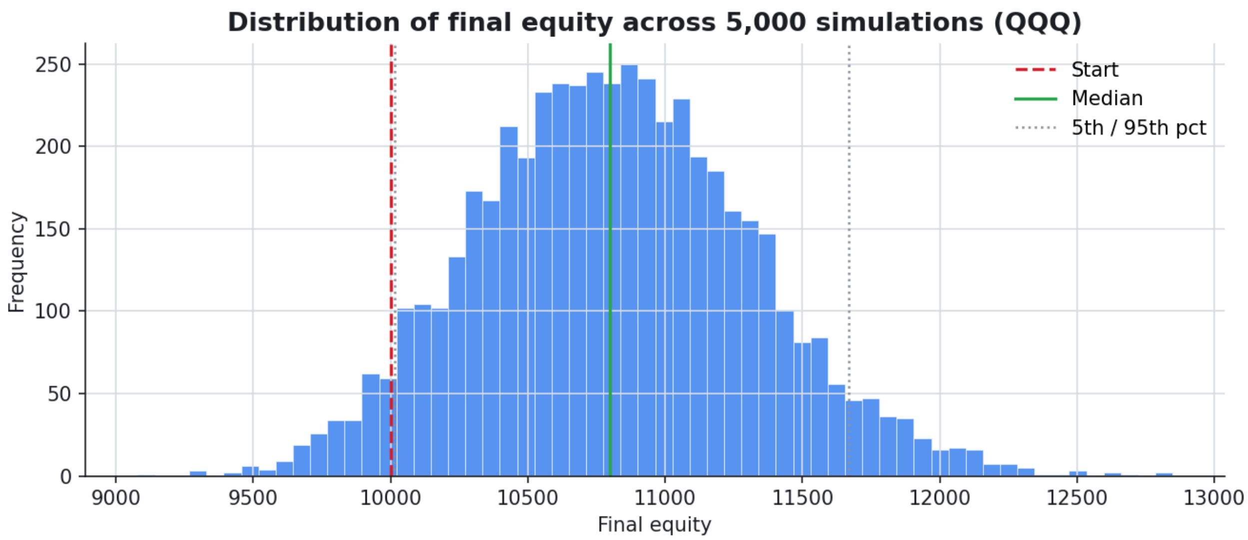
Underwater plot: time spent below the previous equity peak.

4.1 Variance — same edge, different luck

Resampling the realized trades *with replacement* 5,000 times shows the range of outcomes the same system could have produced. (A plain reshuffle would leave the final equity unchanged under fixed-fractional sizing — a product commutes — so resampling with replacement is what disperses the destination, which is exactly what the histogram shows.) The median is the honest expectation; the 5–95% band is the experience you must be able to sit through.

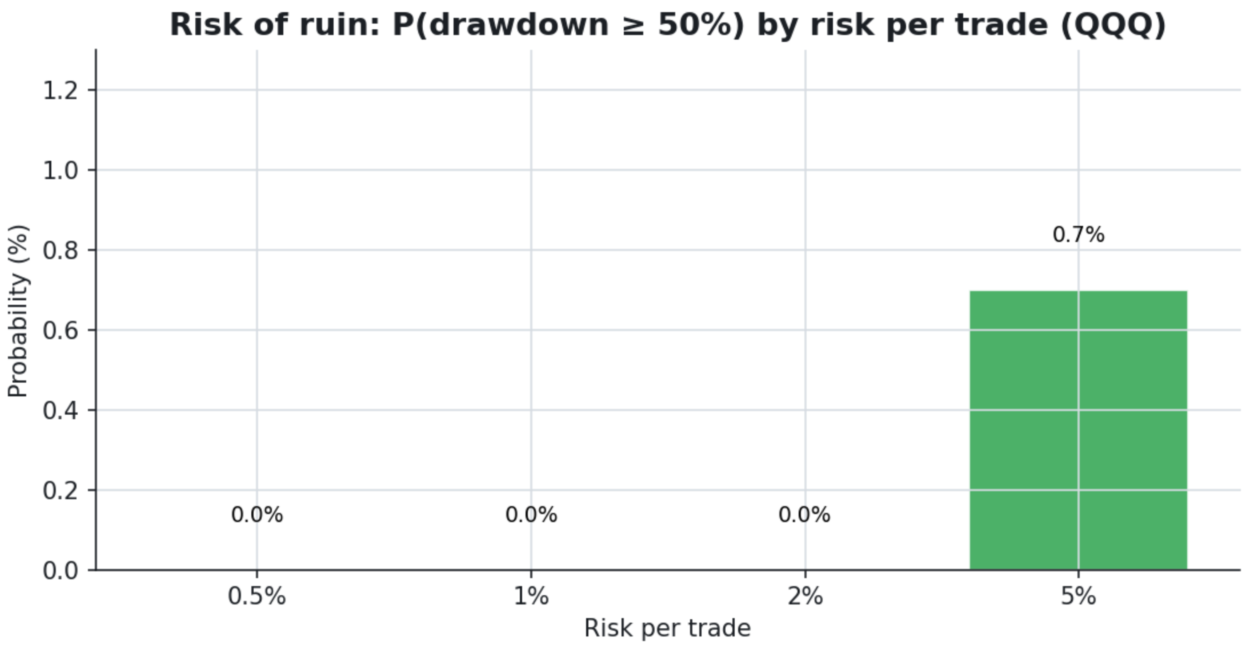


Monte-Carlo fan: median path and 5–95% band vs the single realized run.

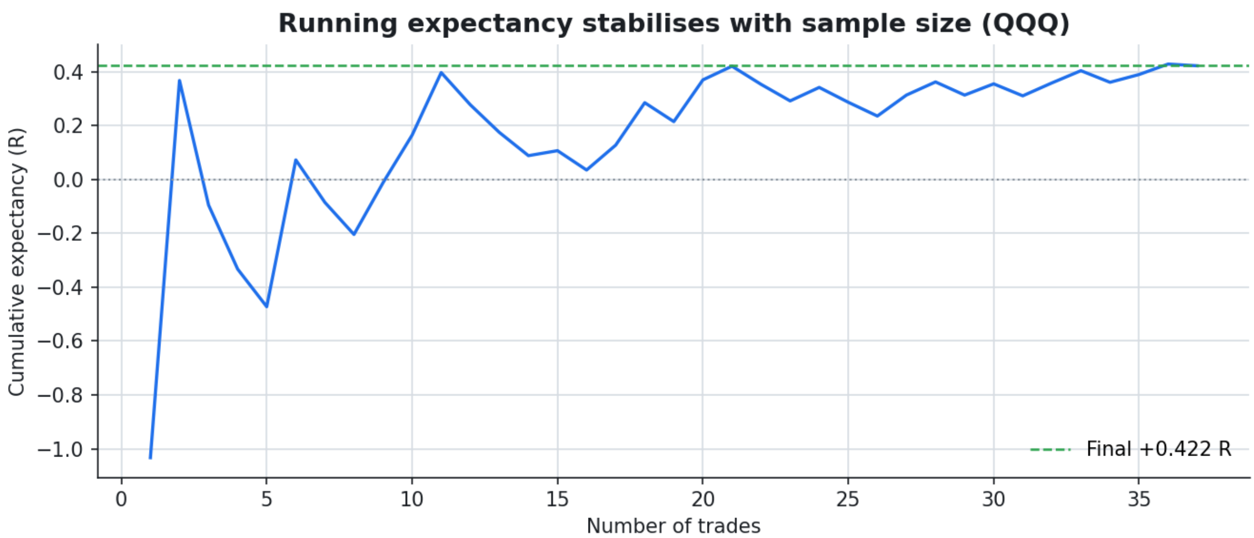


Distribution of final equity across the simulations.

4.2 Risk of ruin & sample size



Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.

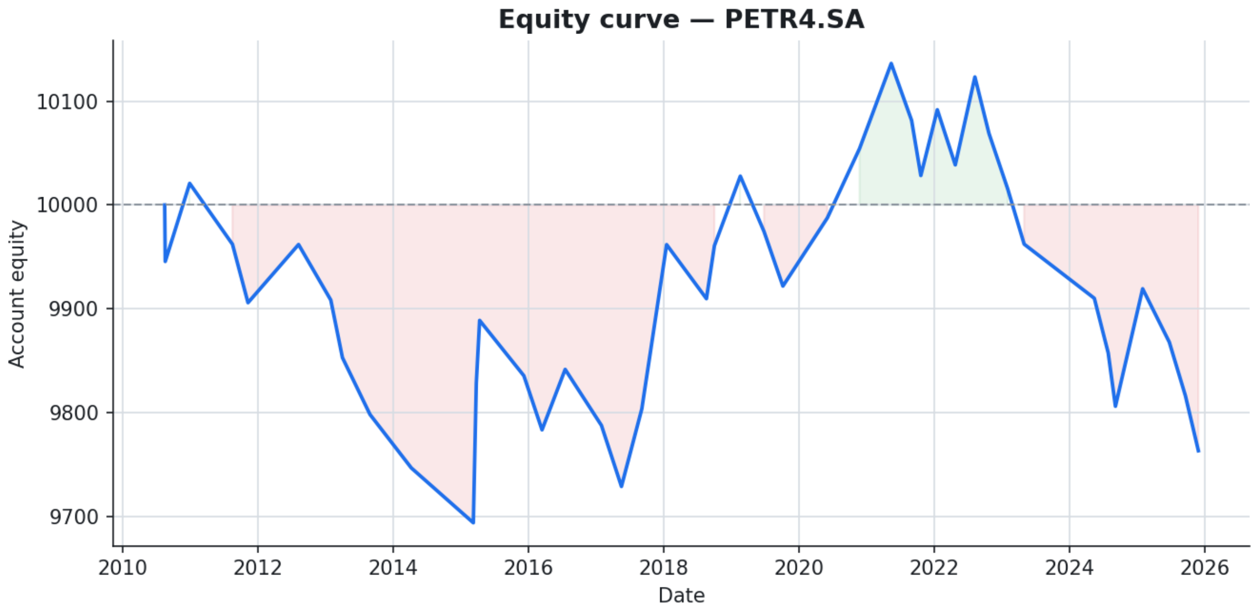


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

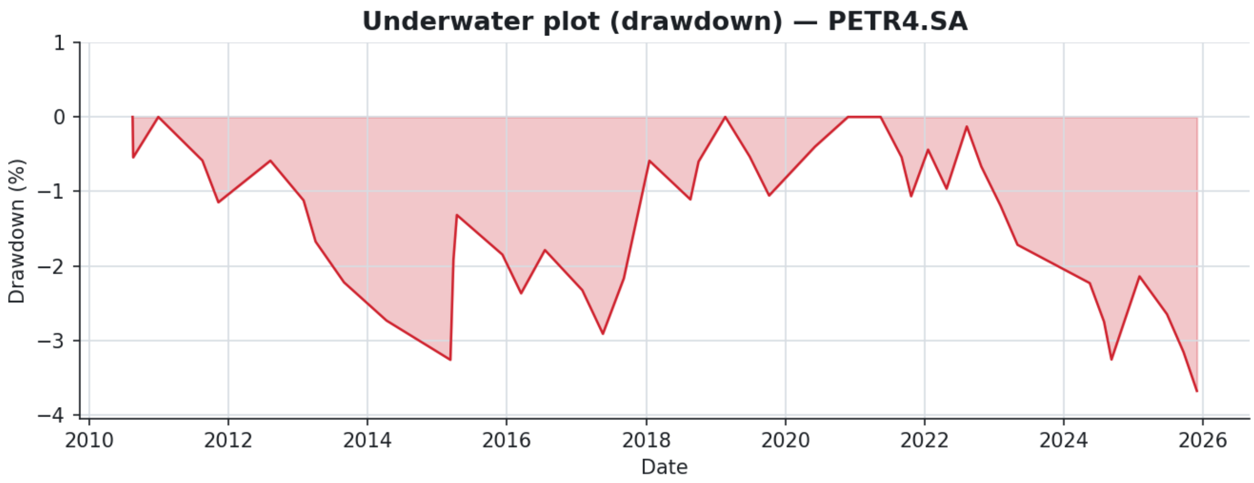
5. PETR4.SA — the full fingerprint

Metric	Value	Metric	Value
Trades	42	**Profit factor**	0.84
Win rate	35.71%	**Payoff (R)**	1.52
Avg win	80.71 (+1.63R)	**Avg loss**	-53.60 (-1.08R)
Expectancy $	-5.63	**Expectancy R**	-0.109
Breakeven WR	39.75%	**Actual WR**	35.71%
Total return	-2.37%	**CAGR**	-0.16%
Max drawdown	3.68%	**Sharpe / Sortino**	-0.14 / -4.86

■ Small sample (42 trades < 100): expectancy is still noise-dominated. Treat with skepticism. In particular, **Sharpe / Sortino and the risk-of-ruin percentages are unstable at this n** — their decimals carry false precision and should be read as rough indications only; the significance and pooling sections (§3-4) are where the conclusions live.



Realized equity curve, costs included.



Underwater plot: time spent below the previous equity peak.

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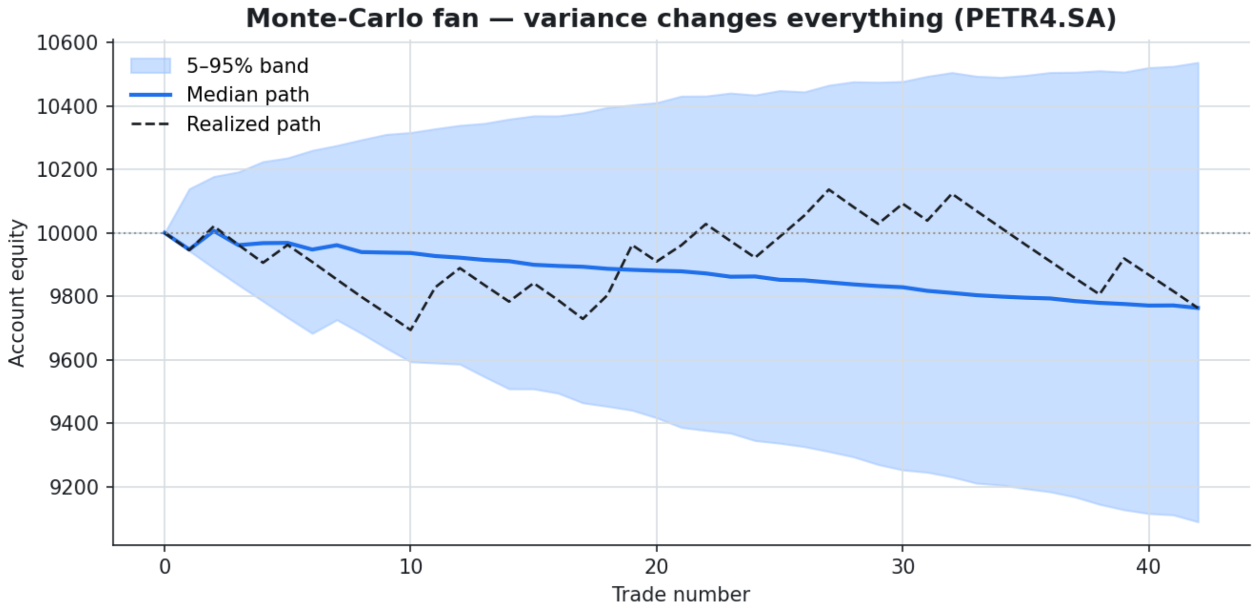

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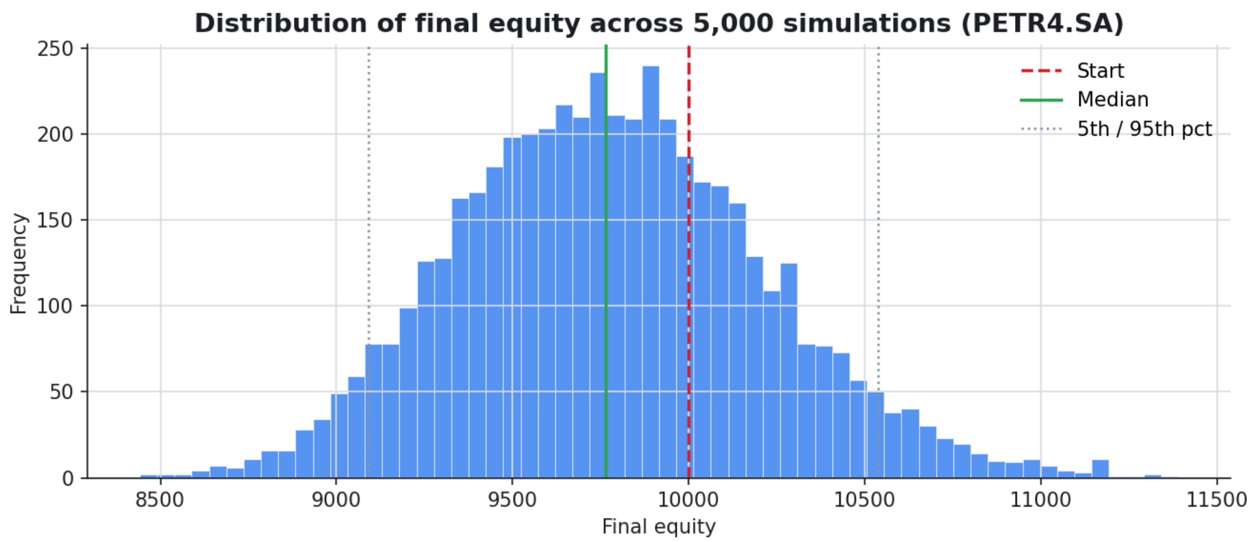

Distribution of final equity across the simulations.

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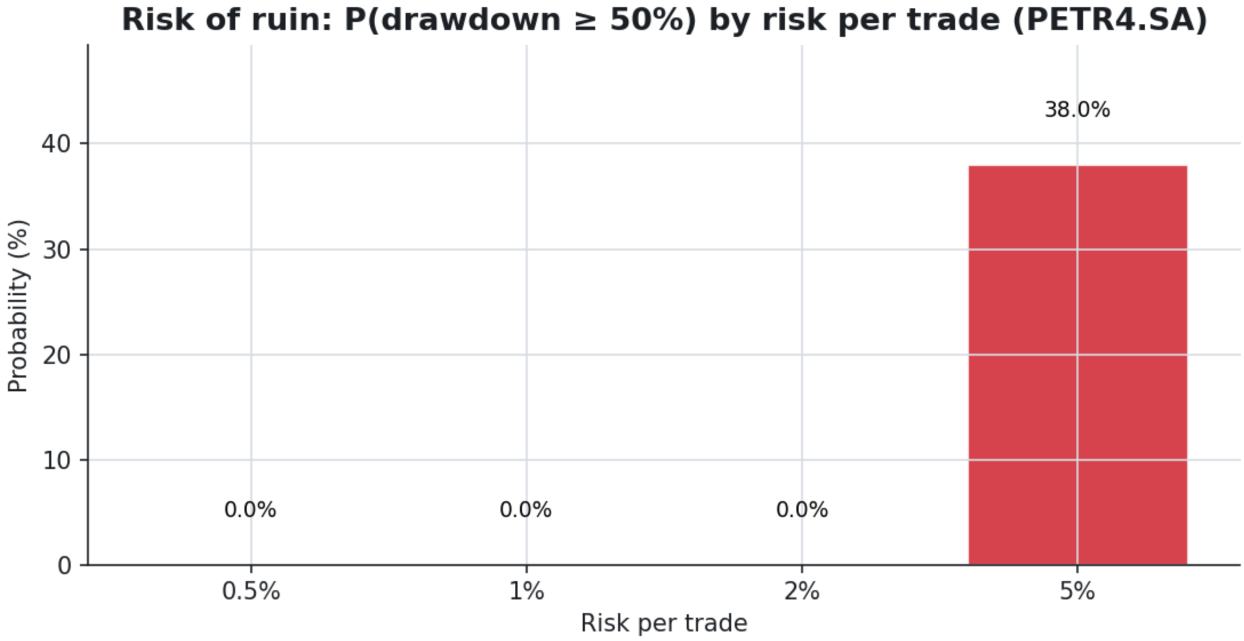

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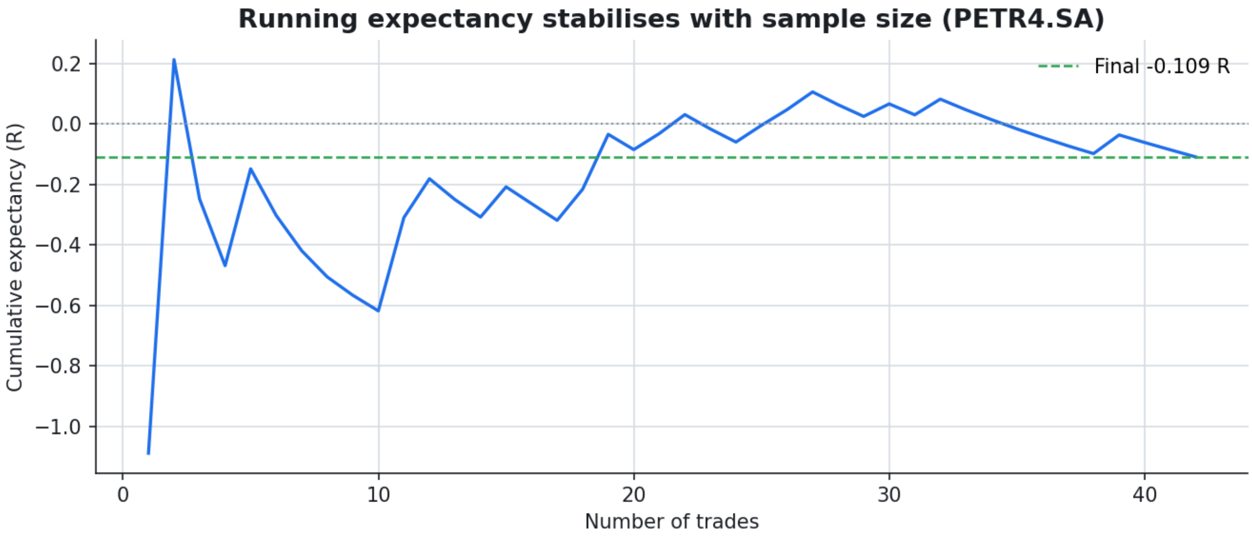
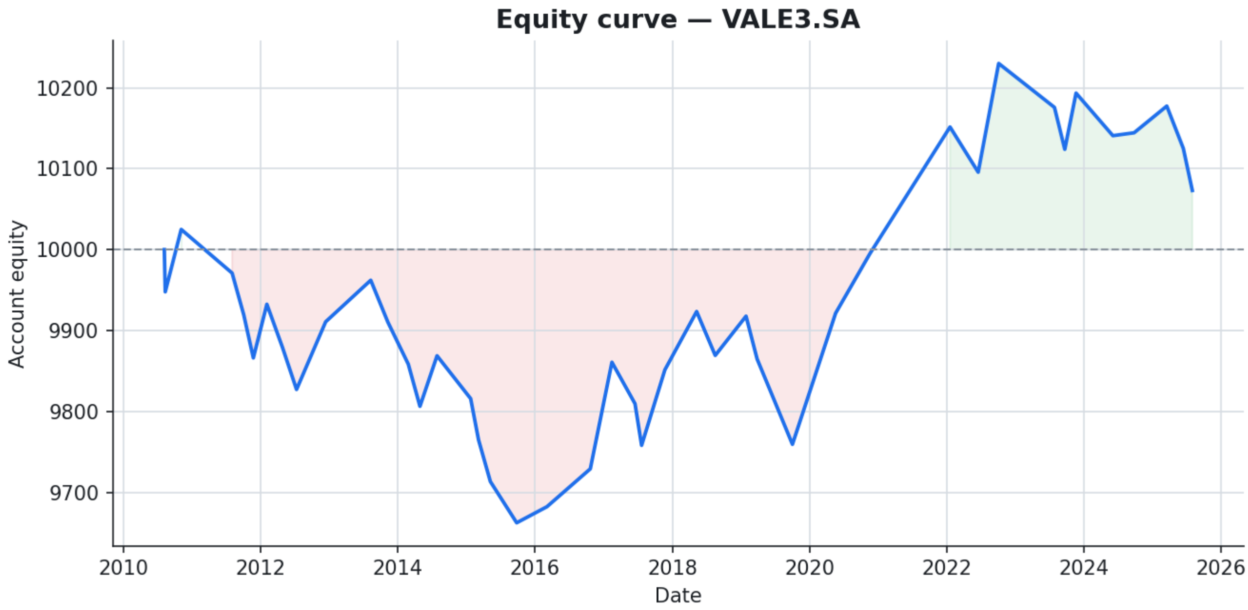


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

6. VALE3.SA — the full fingerprint

Metric	Value	Metric	Value
Trades	43	**Profit factor**	1.06
Win rate	41.86%	**Payoff (R)**	1.48
Avg win	77.00 (+1.56R)	**Avg loss**	-52.51 (-1.06R)
Expectancy $	1.70	**Expectancy R**	+0.039
Breakeven WR	40.37%	**Actual WR**	41.86%
Total return	+0.73%	**CAGR**	+0.05%
Max drawdown	3.62%	**Sharpe / Sortino**	0.04 / 2.40

■ Small sample (43 trades < 100): expectancy is still noise-dominated. Treat with skepticism. In particular, **Sharpe / Sortino and the risk-of-ruin percentages are unstable at this n** — their decimals carry false precision and should be read as rough indications only; the significance and pooling sections (§3-4) are where the conclusions live.



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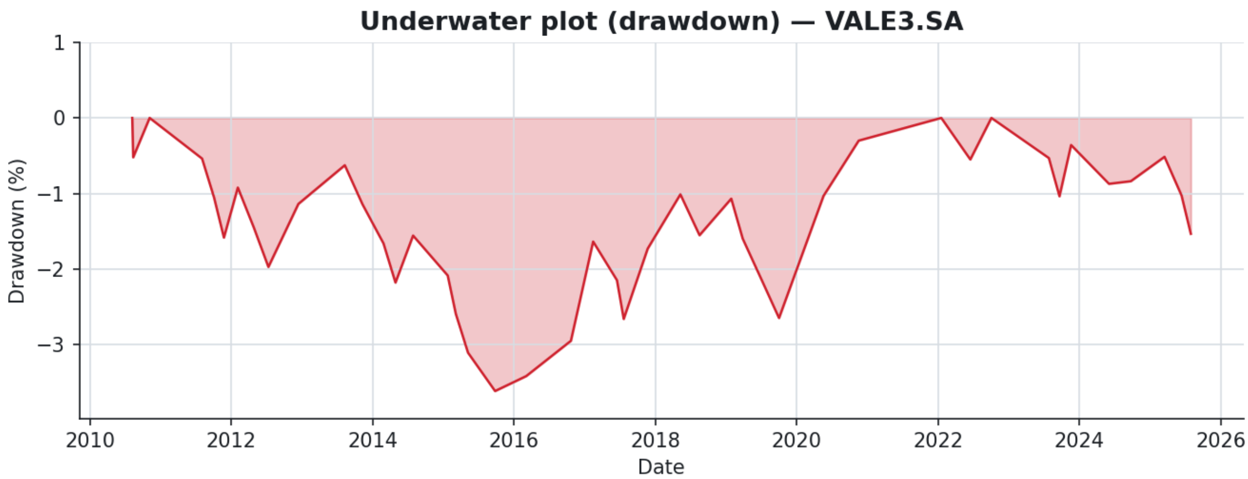

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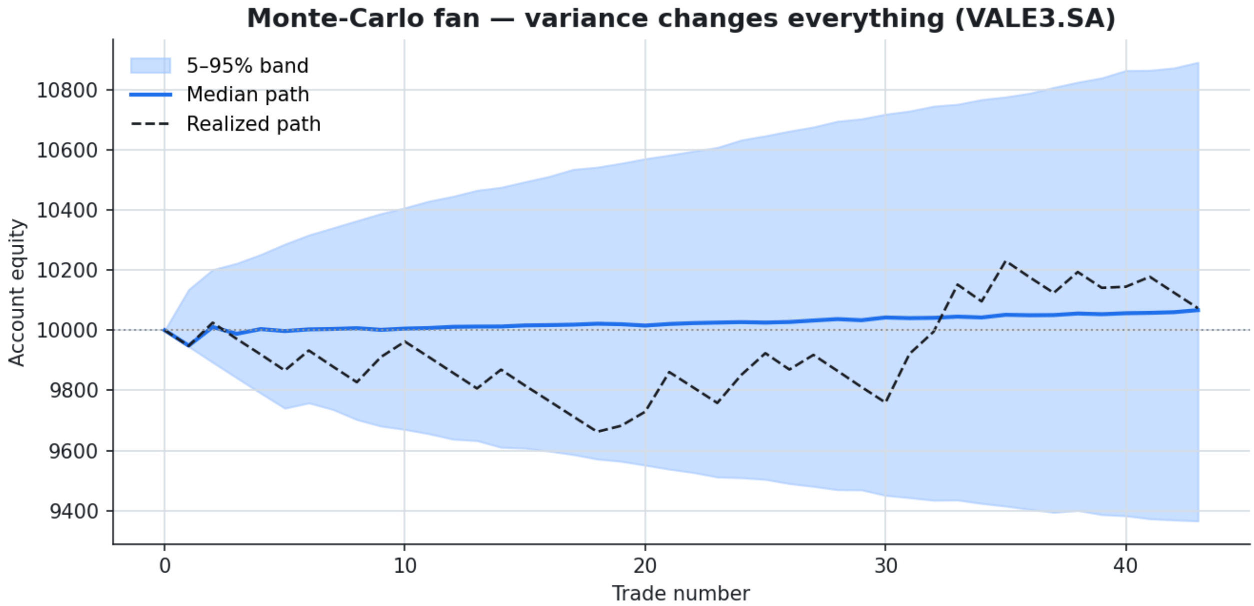

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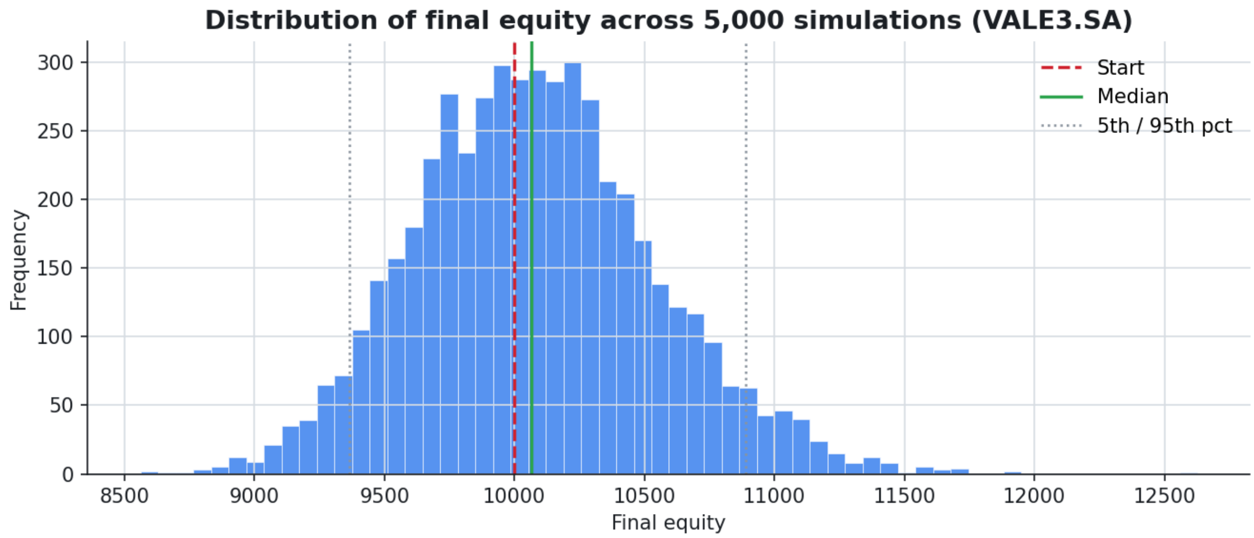

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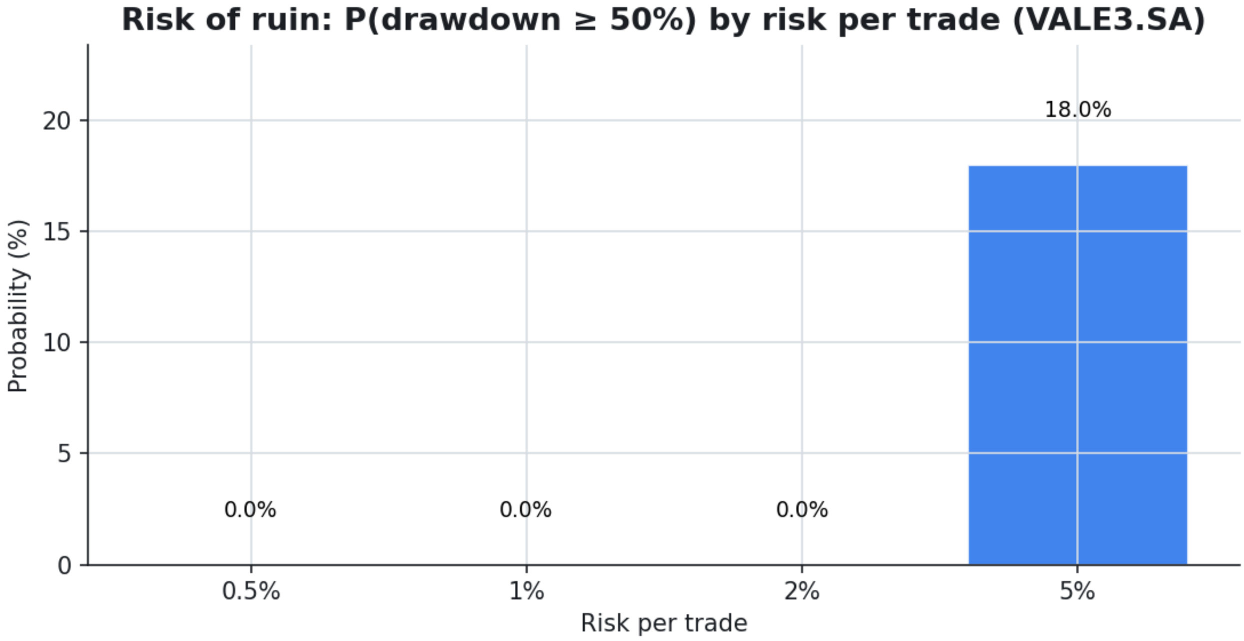

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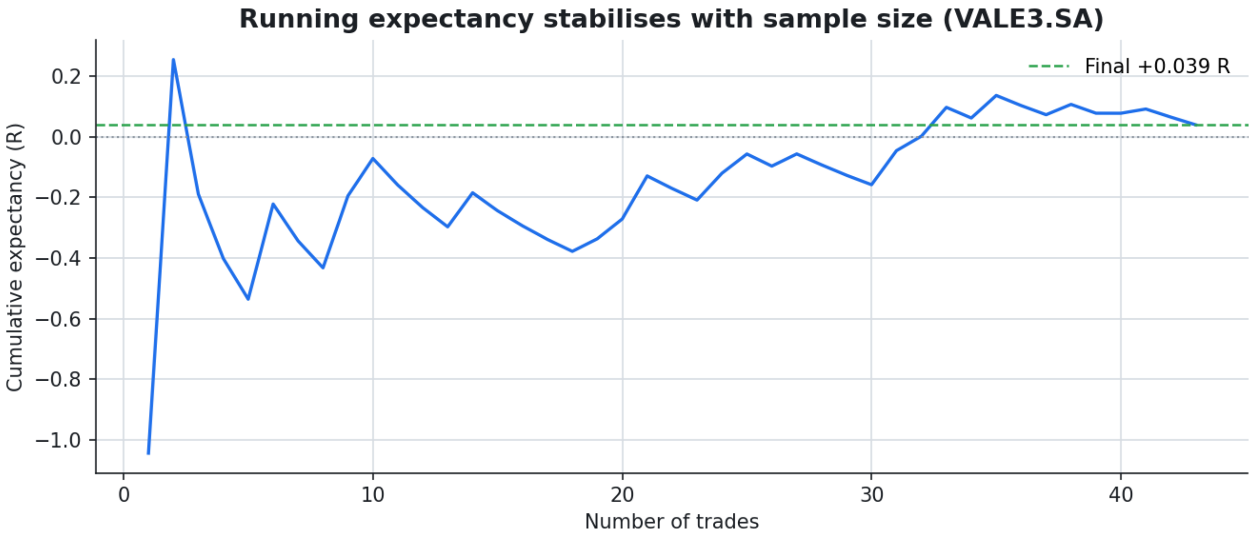
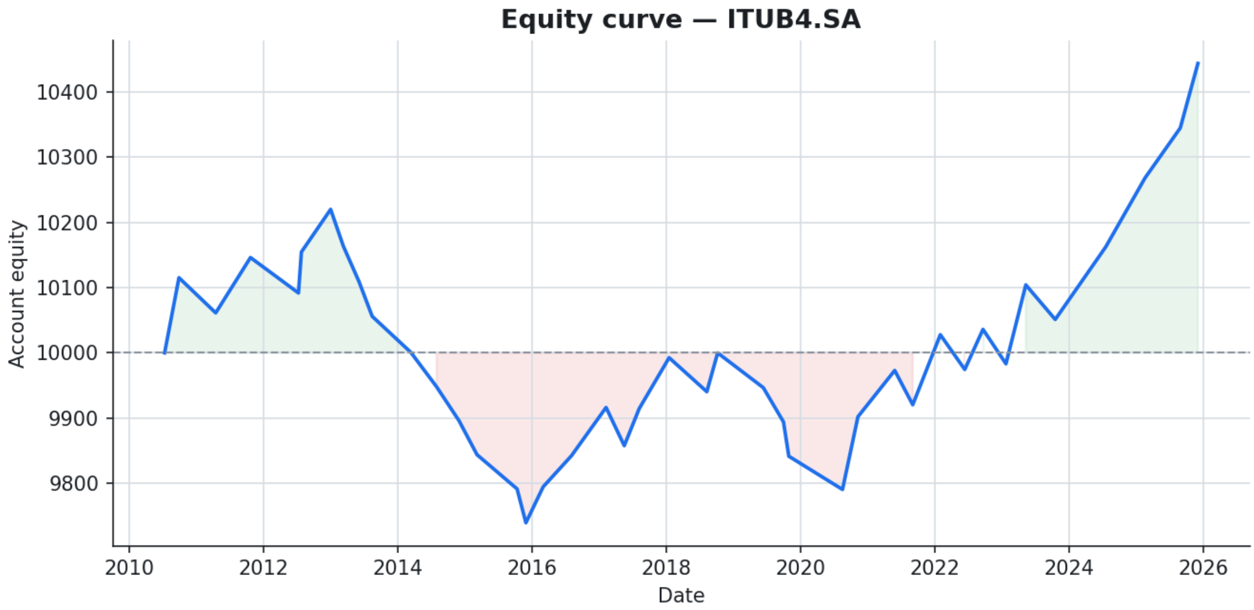


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

7. ITUB4.SA — the full fingerprint

Metric	Value	Metric	Value
Trades	40	**Profit factor**	1.40
Win rate	47.50%	**Payoff (R)**	1.54
Avg win	82.28 (+1.65R)	**Avg loss**	-53.33 (-1.07R)
Expectancy $	11.09	**Expectancy R**	+0.222
Breakeven WR	39.32%	**Actual WR**	47.50%
Total return	+4.43%	**CAGR**	+0.28%
Max drawdown	4.70%	**Sharpe / Sortino**	0.25 / 10.54

■ Small sample (40 trades < 100): expectancy is still noise-dominated. Treat with skepticism. In particular, **Sharpe / Sortino and the risk-of-ruin percentages are unstable at this n** — their decimals carry false precision and should be read as rough indications only; the significance and pooling sections (§3-4) are where the conclusions live.



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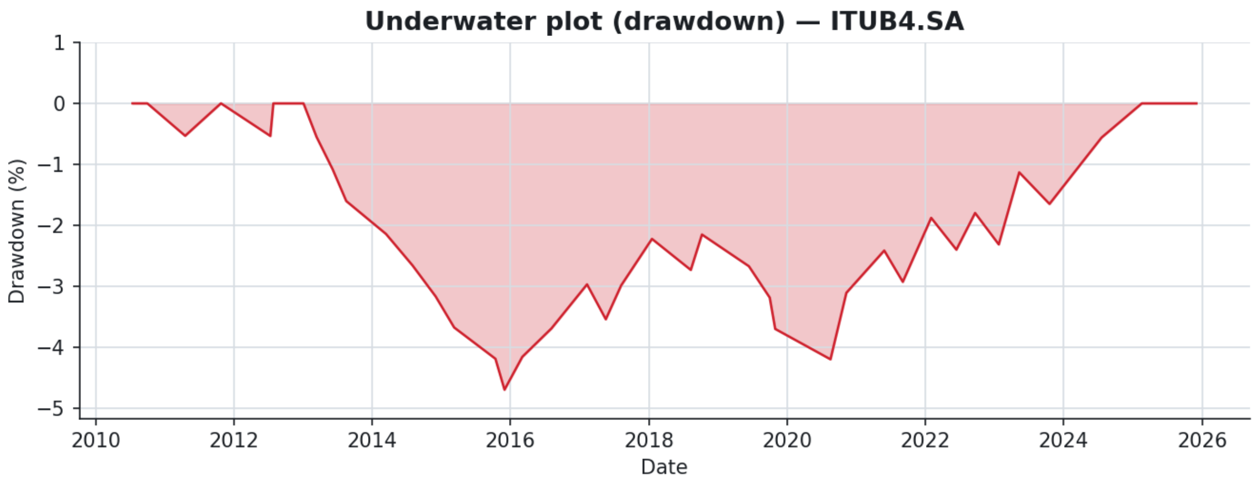

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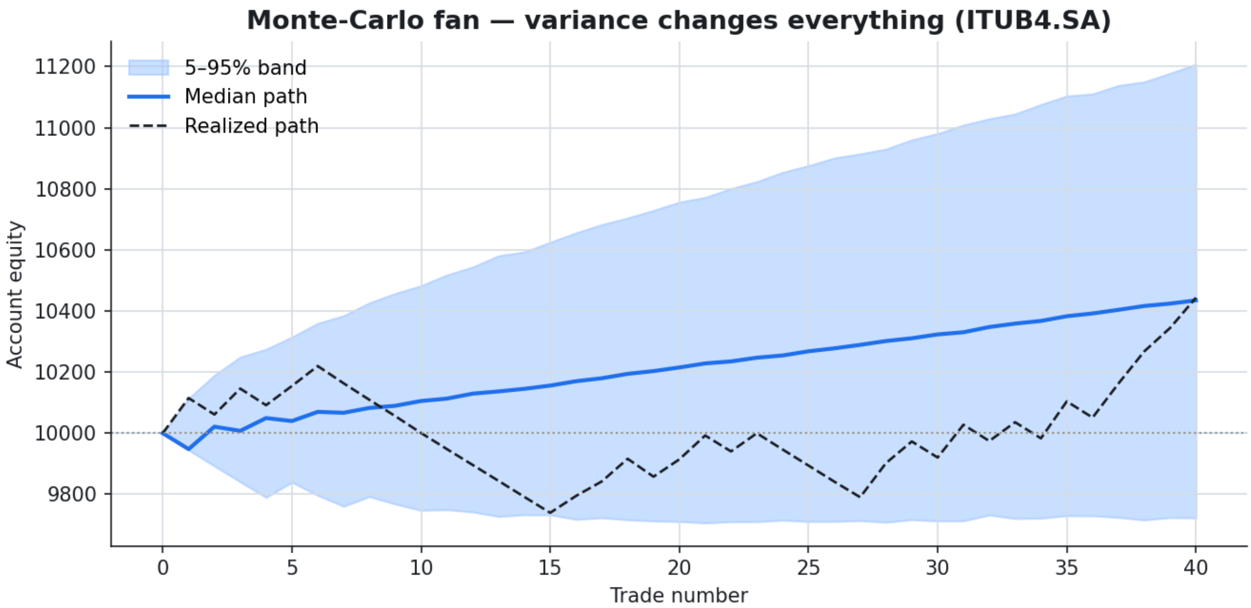

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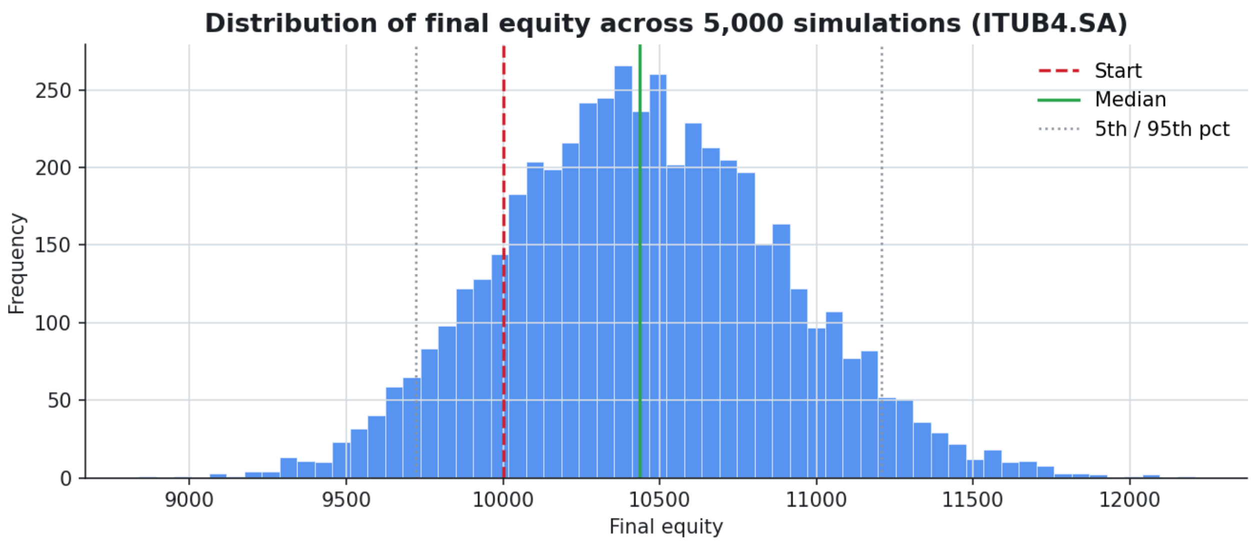

Distribution of final equity across the simulations.

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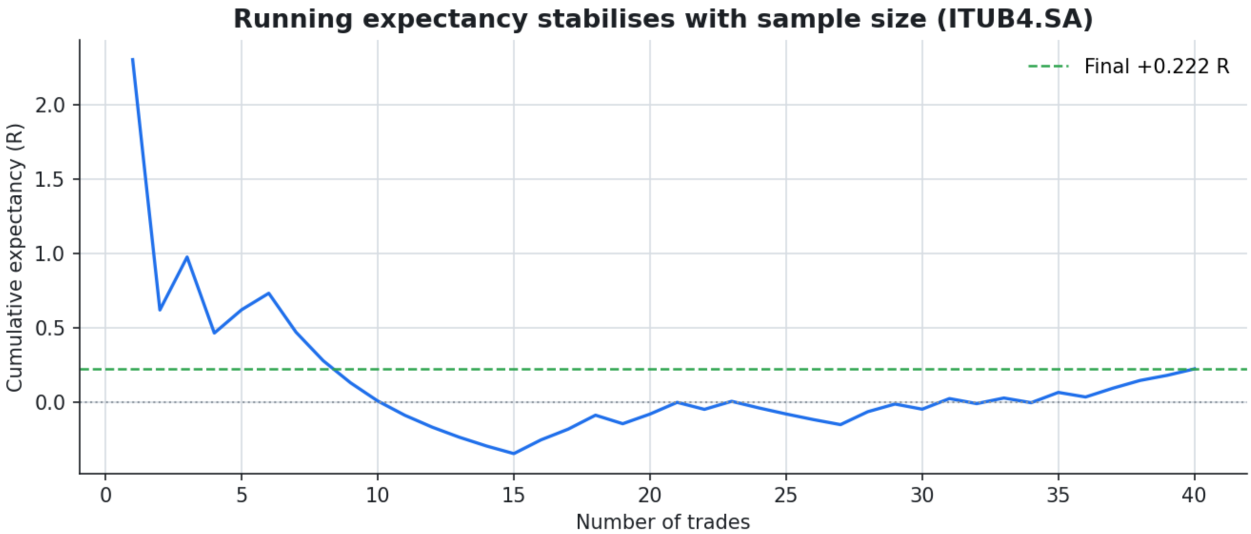
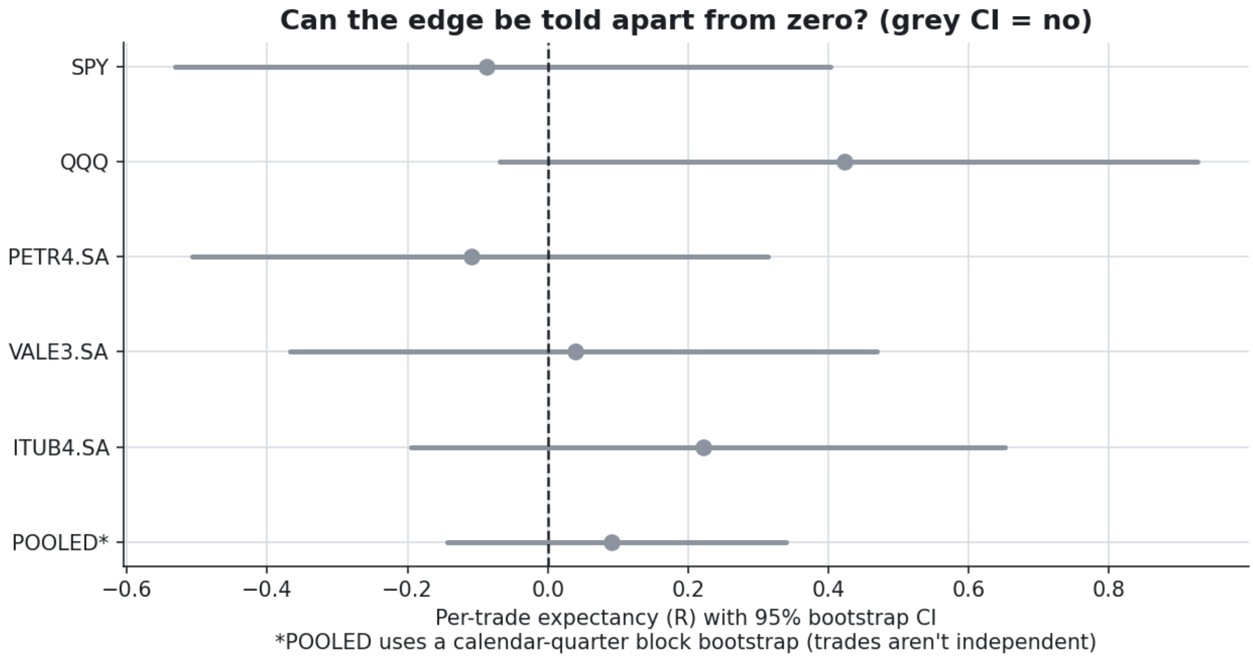


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

8. Is the edge real? Significance, pooling and cost

A point estimate without an interval is a guess with a confident voice. Bootstrapping the per-trade expectancy gives a 95% confidence interval; where it straddles zero, a small edge cannot be told apart from no edge at all.



Expectancy with its 95% bootstrap CI. A grey interval crosses zero — undecidable at this sample size.

Instrument	Trades	Expectancy R	95% CI (R)	P(>0)	Verdict
SPY	39	-0.088	[-0.531, +0.402]	34%	indistinguishable from zero
QQQ	37	+0.422	[-0.068, +0.926]	95%	indistinguishable from zero
PETR4.SA	42	-0.109	[-0.507, +0.314]	30%	indistinguishable from zero
VALE3.SA	43	+0.039	[-0.368, +0.468]	57%	indistinguishable from zero
ITUB4.SA	40	+0.222	[-0.196, +0.651]	85%	indistinguishable from zero
POOLED†	**201**	**+0.090**	**[-0.143, +0.340]**	**77%**	**indistinguishable from zero**

† The POOLED row uses the calendar-quarter block bootstrap [-0.143, +0.340] (the honest interval, since trades aren't independent), matching the forest plot. The i.i.d. interval would be the tighter [-0.106, +0.295].

Of the 5 instruments, **5** have a confidence interval that includes zero: individually, the data cannot confirm an edge in either direction. This is the honest reading the headline scorecard hides.

Two caveats make the picture even more sober. **QQQ sits exactly on the boundary** (P(edge > 0) = 95%), so calling it positive would be a one-tailed knife-edge, not a finding. And **multiple comparisons** bite: testing five instruments, the chance that at least one clears a 95% one-sided bar by luck alone is $1 - 0.95^5 \approx 23\%$. Finding one borderline name among five is roughly what pure noise predicts — another reason not to single QQQ out.

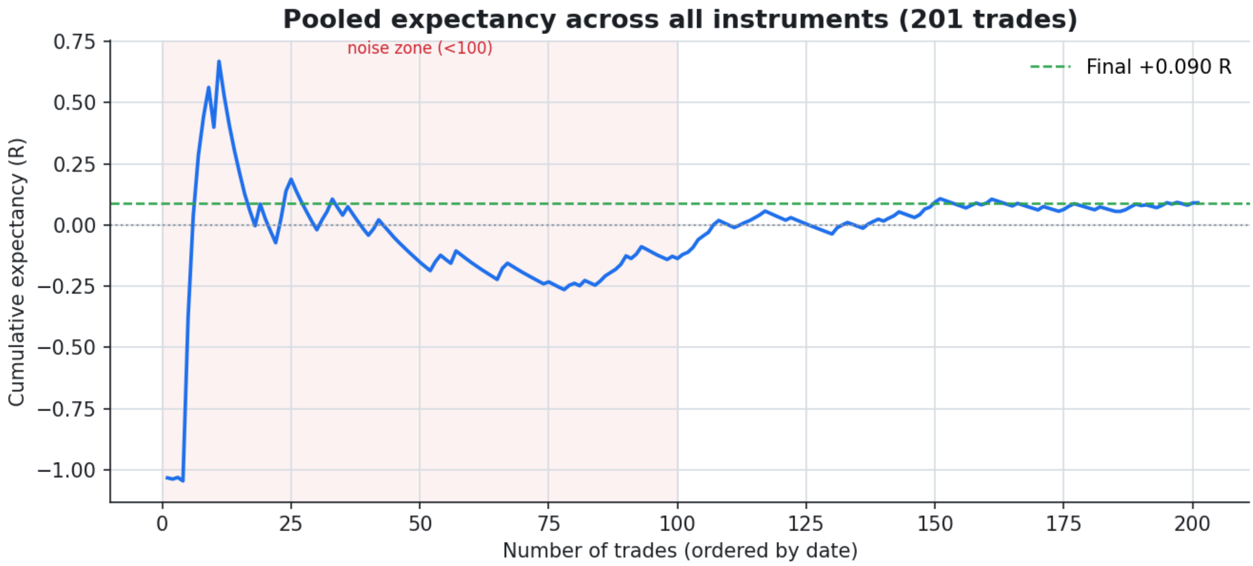
8.1 Pooling for power, and an out-of-sample split

Because R normalises each trade by its risk, the 201 trades from all instruments can be pooled into one stream — finally crossing the ~100-trade threshold. The pooled expectancy is **+0.090R**. But these trades are *not* independent — US indices (~0.9 correlated) and the Brazilian names move together, so trades firing in the same period carry redundant information. An i.i.d. bootstrap would understate the uncertainty; a **calendar-quarter block bootstrap** keeps that correlation intact and gives the honest interval:

• i.i.d. CI (optimistic): [-0.106, +0.295]R
• block CI (58 quarters, honest): **[-0.143, +0.340]R** — still includes zero (P(edge > 0) = 77%)

Splitting the pool chronologically into halves gives a genuine out-of-sample check:

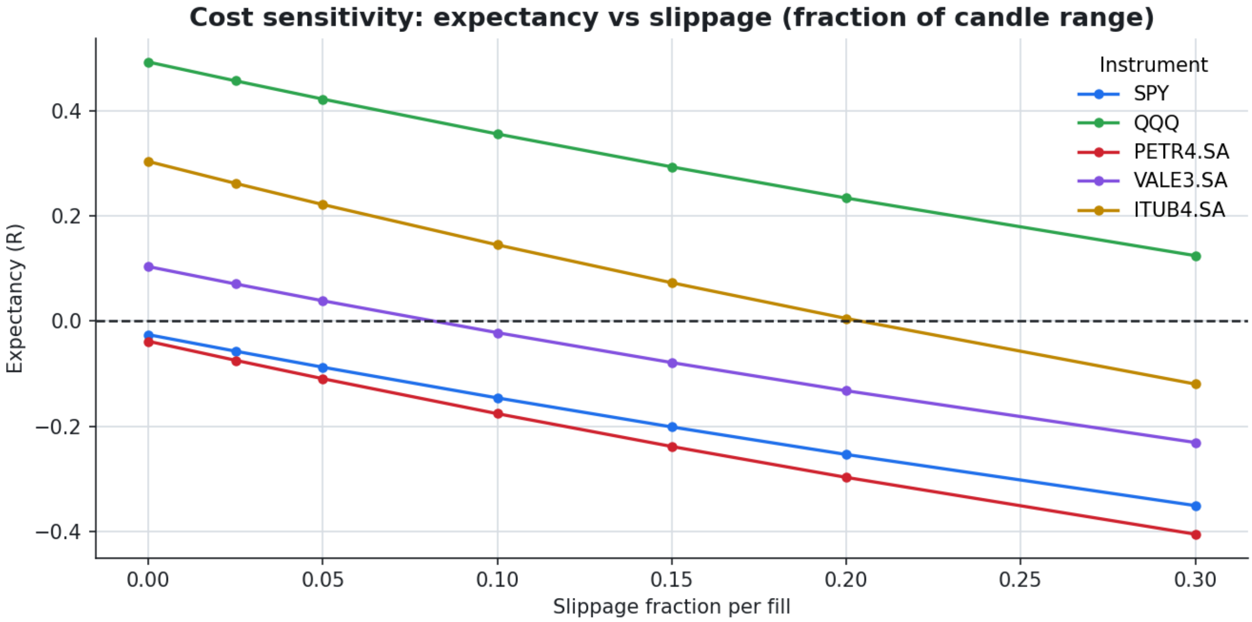
• In-sample (first 100 trades): **-0.137R**
• Out-of-sample (last 101 trades): **+0.316R**



Pooled running expectancy. Even at ~200 trades it is still settling; the shaded zone marks the sub-100 region where the estimate is noise.

8.2 The edge is thin: cost sensitivity

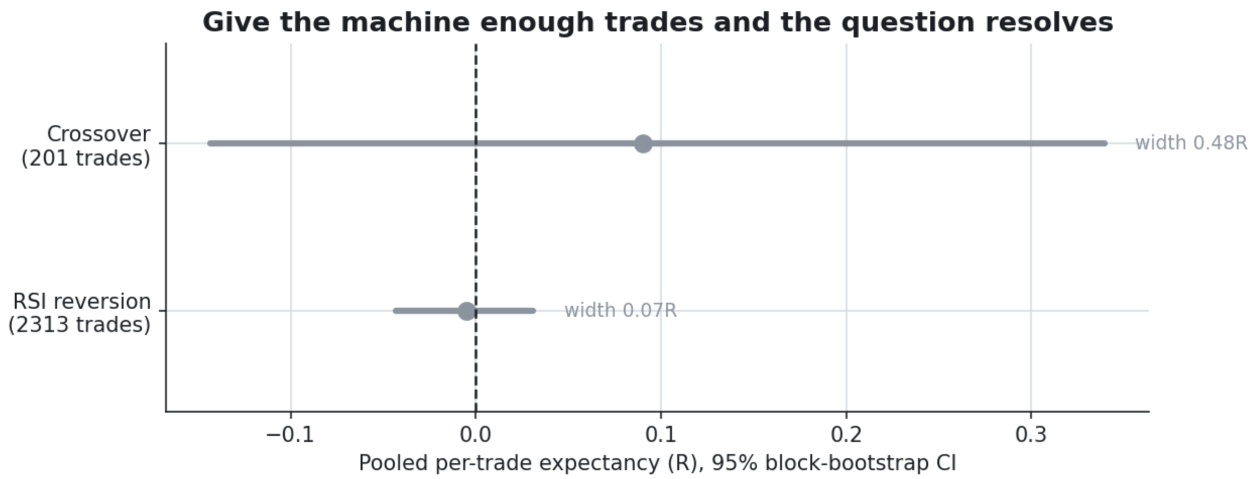
Where expectancy is a fraction of an R, the cost assumption is a lever, not a footnote. Sweeping the slippage assumption shows how fast each instrument's edge crosses into the red.



Expectancy vs slippage. Lines that dip below the dashed zero line have lost their edge to costs — for the marginal names, that happens with a small change in assumptions.

9. Giving the machine enough trades

Every limitation above traces back to one root cause: a daily moving-average crossover fires ~2.5 times a year, so even 16 years cannot power the test. The fix is not statistical, it is the sample. Swapping in a frequent strategy — an RSI(2) mean-reversion that buys short-term dips inside uptrends — over a basket of 20 instruments produces **2,313 trades**. The same engine, the same maths; only the setup changed.



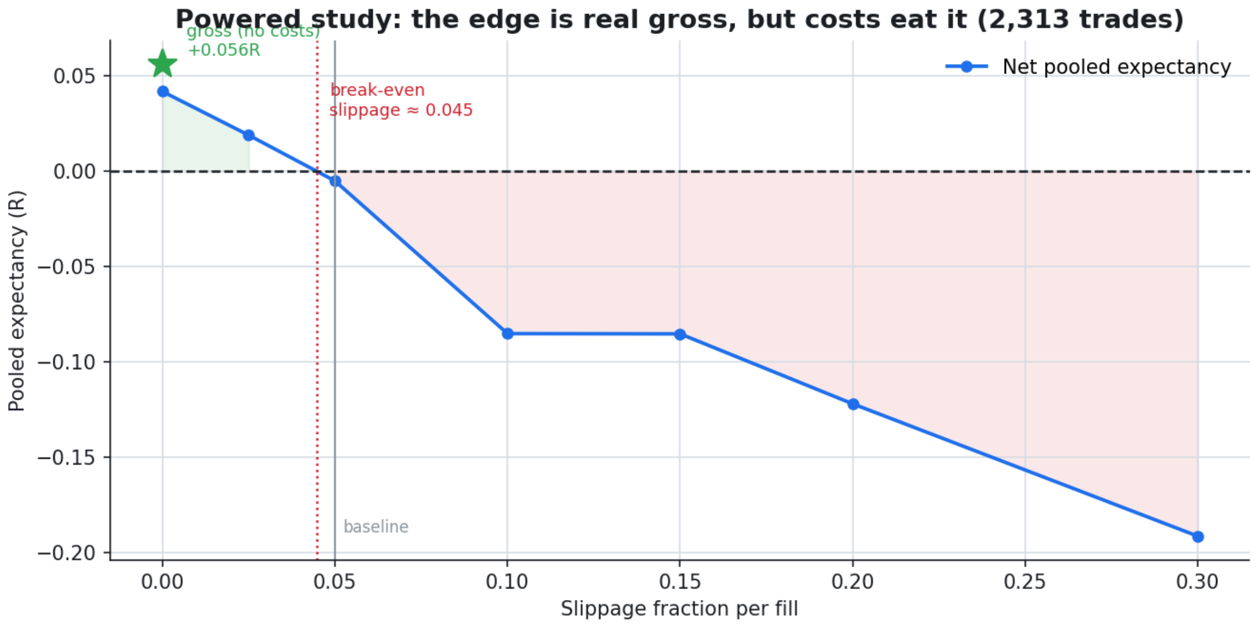
The pooled 95% block-bootstrap CI collapses from 0.48R wide (crossover, undecidable) to 0.07R wide (reversion) — a precise band on zero.

The powered pooled expectancy is **-0.005R**, block CI **[-0.043, +0.031]**. The verdict is no longer "we cannot tell" — it is the sharper "the edge is, to a tight tolerance, zero after costs." The high 60-68% win rates on US indices are the genuine mean-reversion signature, but the small per-trade gains are eaten by spread and slippage. In-sample (-0.001R) and out-of-sample (-0.009R) agree, so this is stable, not a fluke.

A note on the basket. These 20 names are liquid blue chips that exist *today*, so the selection is survivor-biased. Crucially, that bias works *in favour* of finding an edge (survivors are the winners) — and the result is still zero. The null finding is therefore conservative: a survivorship-free basket would, if anything, look worse.

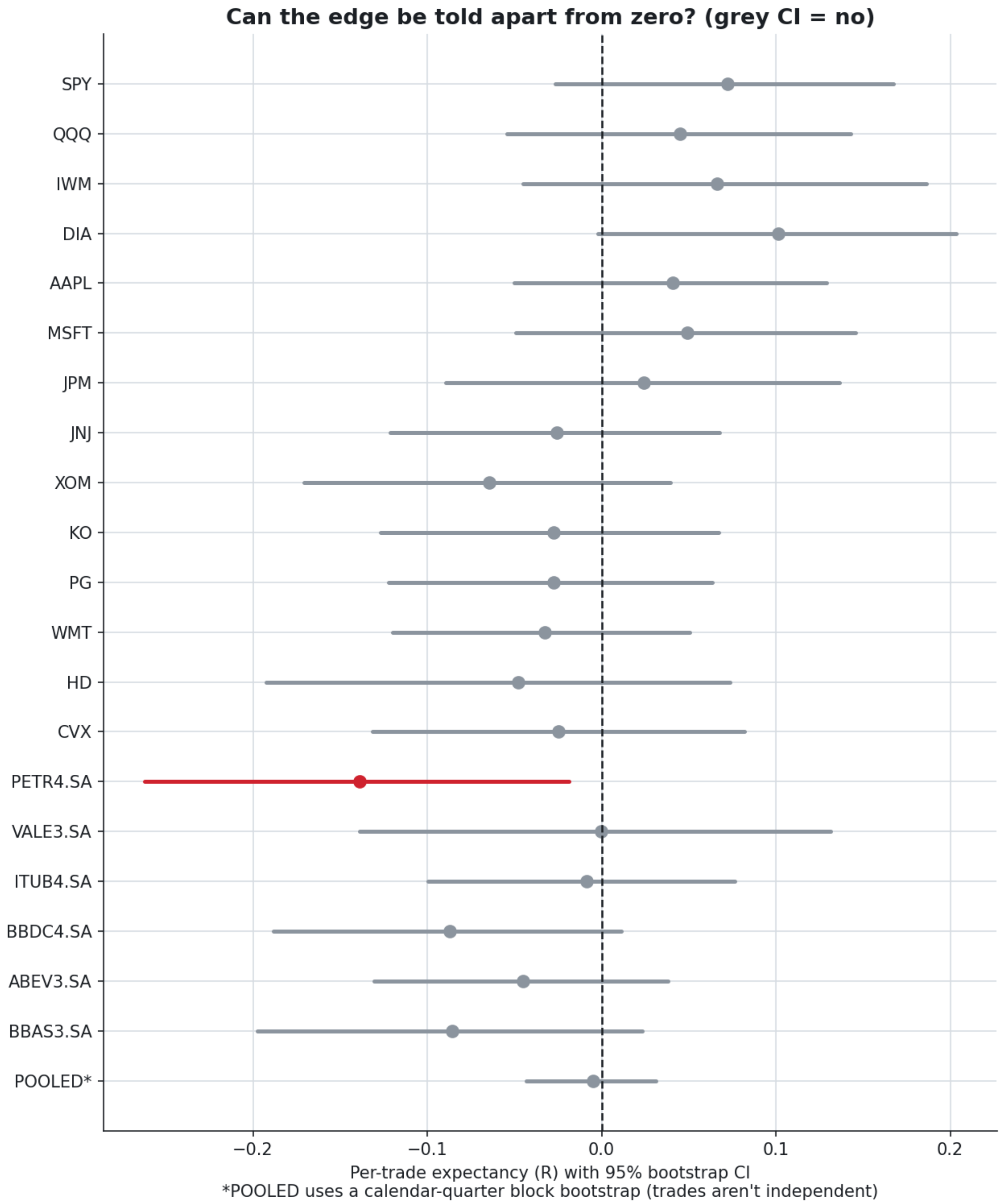
9.1 Gross vs net — where the edge dies

The "zero after costs" verdict rests on the cost assumption, so here it is stressed on the study that depends on it. With **all frictions removed**, the pooled signal is **+0.056R** — genuinely positive. It is the costs that erase it: the pooled edge crosses zero at a break-even slippage of ≈ **0.045 of the candle range**, right around the realistic baseline. The reading is precise and practical: the mean-reversion signal is *real but small*, and only a low-cost (institutional) participant operating below that break-even would keep it; at retail frictions it nets to nothing.

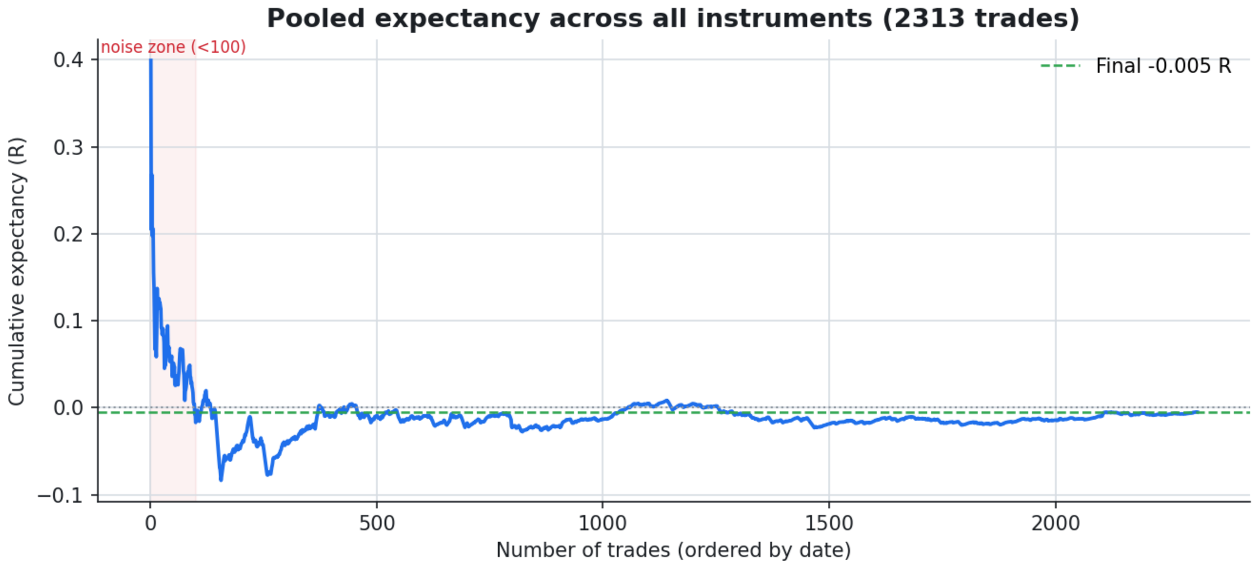


Pooled expectancy vs slippage. The green star is the gross signal; the edge is positive until the break-even slippage, then negative. Costs, not the signal, decide the verdict.

9.2 Per-instrument and convergence



With ~100+ trades each the intervals tighten; PETR4 even turns significantly negative. The pooled band sits precisely on zero.



Pooled running expectancy over 2,313 trades — it settles smoothly near zero, the calm that only sample size buys.

10. The recovery math — why protecting capital wins

Losses and the gains needed to undo them are asymmetric. A 50% drawdown does not need 50% to recover — it needs 100%. This is why risk of ruin, not raw return, is the metric that keeps a trader in the game.

Drawdown suffered	Gain required to recover
10%	11%
20%	25%
30%	43%
40%	67%
50%	100%

Disclaimer

This is an educational and research project. Nothing here is investment advice. The example strategy is a didactic moving-average crossover, not a system with a guaranteed edge. The backtester *measures* an edge; it does not create one. Past performance does not guarantee future results.